Detecting AI-Generated Images via CNN Backbone Architectures with Texture Feature Fusion, Patch Loss, and Contrastive Learning

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Abstract

The proliferation of AI-generated imagery from generative models such as Stable Diffusion, Midjourney, and GANs poses a growing threat to digital media authenticity. This paper presents a systematic empirical study of five CNN-based architectures for binary AI-generated vs. real image classification on a large-scale, class-imbalanced dataset of 180,477 images. Starting from a plain deep CNN baseline trained with binary cross-entropy, we progressively introduce residual connections, focal loss, Gray-Level Co-occurrence Matrix (GLCM) texture feature fusion, patch-level supervision, and contrastive loss. We report imbalance-aware metrics including F1-score, PR-AUC, ROC-AUC, Balanced Accuracy, and Matthews Correlation Coefficient (MCC) across all five models. Our best-performing fully trained model — a Residual CNN with Focal Loss — achieves an ROC-AUC of **0.8075** and an F1-score of **0.7095** on the full held-out test set. The GLCM-fused tandem CNN achieves the highest F1 of **0.7616** on its evaluation split, underscoring the benefit of statistical texture features. We also demonstrate that patch-level and contrastive auxiliary objectives are promising but require extended training to converge. Our work contributes a reproducible, multi-metric evaluation protocol, Grad-CAM interpretability analysis, and a public benchmark dataset.

1 Introduction

The rapid advancement of text-to-image and video synthesis models — including Stable Diffusion [1], Midjourney, and DALL-E — has made AI-generated imagery visually indistinguishable from authentic photographs in many cases. A 2025 human-study benchmark reported that humans detect AI-generated content at near coin-toss accuracy [2], underscoring the urgency of automated detection.

*Dataset: `ShreyashDhoot/AI_vs_Real` on Hugging Face Hub.

Automated AI-generated image detection lies at the intersection of image forensics, computer vision, and trustworthy AI. Prior work has ranged from frequency-domain analysis [3] and GAN fingerprint extraction [4] to vision transformer-based probing [7] and multi-modal texture fusion [8]. However, many existing approaches: (i) rely on pretrained large backbones that are computationally expensive, (ii) do not report imbalance-aware metrics critical for real-world deployment, and (iii) focus on a single architecture rather than a systematic progression of design choices.

This paper fills this gap by presenting five compact CNN architectures evaluated under a consistent and rigorous protocol on a large-scale dataset. Our contributions are:

1. A systematic progression from plain CNN to residual+focal loss, to GLCM-augmented CNN, to patch-supervised CNN, and contrastive CNN — each ablating a specific design decision.
2. Extensive imbalance-aware evaluation including F1, PR-AUC, ROC-AUC, Balanced Accuracy, MCC, and optimal threshold selection.
3. Integration of GLCM handcrafted texture features with a deep CNN backbone for AI-generated image detection on a large-scale dataset.
4. Application of patch-level auxiliary loss and margin-based contrastive loss in this task domain.
5. Grad-CAM visualizations establishing interpretability of what spatial features drive classification decisions.
6. A publicly available Hugging Face dataset (`ShreyashDhoot/AI_vs_Real`) for future benchmarking.

2 Related Work

2.1 Frequency and Spectral Methods

Early work on detecting GAN-generated images exploited spectral artifacts. Frank et al. [3] showed that images generated by upsampling-based generators exhibit characteristic checkerboard artifacts in the DCT frequency domain. Similarly, Durall et al. [5] analyzed the radial power spectrum of GAN outputs, finding that generators fail to reproduce natural image frequency statistics. These approaches are effective against specific generators but generalize poorly to diffusion-based synthesis, which produces more naturalistic frequency distributions.

2.2 CNN-Based Forensic Detectors

Wang et al. [4] demonstrated that a ResNet-50 classifier trained on ProGAN-generated images generalizes surprisingly well to unseen GAN architectures, establishing the CNNDetect benchmark. Gragnaniello et al. [6] extended this analysis to Latent Diffusion Models, finding that detectors trained on GAN outputs struggle against diffusion-generated images. More recent work by Corvi et al. [7] systematically benchmarked multiple detectors against several diffusion model families. Ensemble approaches combining ResNet50, DenseNet, and transfer learning have achieved up to 94% accuracy on smaller benchmarks [13].

2.3 Texture-Aware Approaches

Texture statistics have long been used in image forensics. The Gray-Level Co-occurrence Matrix (GLCM) captures second-order pixel intensity co-occurrence statistics — including contrast, correlation, energy, and homogeneity — that reflect the regularity of synthetic textures. A 2025 multi-modal fusion network [8] integrates RGB, Local Binary Patterns (LBP), and GLCM representations in a three-branch architecture, reporting improved detection over single-modality baselines. Our GLCM-fused tandem models are motivated by and align with this direction, extending it to a large-scale 180K-image training setting.

2.4 Patch-Based and Contrastive Methods

The Panoptic Patch Learning (PPL) framework [9] introduces randomized patch reconstruction augmentation combined with patch-wise contrastive learning, operating under the principle that "All Patches Matter." The contrastive image forgery localization method CFL-Net [10] demonstrated that supplementing cross-entropy with contrastive loss substantially improves localization of forged regions. Our Models 4 and 5 draw from these inspirations, applying patch-level supervision and margin-based contrastive objectives within a compact CNN backbone, showing these objectives are feasible without large transformer backbones.

2.5 Class Imbalance in Detection

A largely underexplored aspect in the literature is class imbalance. Focal Loss [11], originally proposed for dense object detection, addresses this by down-weighting easy examples and focusing training on hard misclassified samples. Our work explicitly evaluates focal loss vs. binary cross-entropy under a 1.49:1 fake-to-real imbalance with full metric reporting.

3 Dataset

3.1 Source and Scale

We use the `ShreyashDhoot/AI_vs_Real` dataset [14] hosted on Hugging Face Hub. All images are converted to RGB and resized to 160×160 pixels. The full dataset is split 80/10/10 into training, validation, and test sets.

Table 1: Dataset statistics and class distribution.

Split	**Count**	**Fraction**
Train	144,381	80%
Validation	18,048	10%
Test	18,048	10%
Train: AI-generated (0)	86,393	59.8%
Train: Real (1)	57,988	40.2%

3.2 Class Imbalance

The training split exhibits a 1.49:1 fake-to-real imbalance. This is critical: naive models over-predict the majority class (fake), yielding inflated accuracy but poor recall on real images. We address this via class-weight compensation:

$$w_c = \frac{N_{\text{total}}}{K \cdot N_c} \tag{1}$$

yielding $w_{\text{fake}} = 0.8356$ and $w_{\text{real}} = 1.2449$.

3.3 Evaluation Protocol Note

Models 1–2 (TensorFlow) are evaluated on the full 18,048-sample test split. Tandem models 3–5 (PyTorch) are evaluated on a 250-sample streamed test subset due to computational constraints. Direct numeric comparison across protocols should be treated as indicative only; future work will unify these splits.

4 Architectures

4.1 Model 1: Plain Deep CNN (Baseline)

The baseline architecture `deep_dense_cnn` comprises four sequential convolutional blocks with channel doubling (16→32→64 →128), each followed by Batch Normalization, MaxPooling, and increasing Dropout rates (0.20→0.35). The classification head applies Global Average Pooling, then Dense(128) and Dense(64) with Batch Normalization and Dropout, terminating in a sigmoid output. No residual connections or L2 regularization are used. Total trainable parameters: ≈321K.

The model is trained with Binary Cross-Entropy (BCE) loss and class weights, using Adam ($lr = 10^{-4}$), ReduceLROnPlateau (factor 0.3, patience 2), and EarlyStopping on `val_loss` (patience 5). Best weights restored from epoch 14 of 19 trained epochs.

4.2 Model 2: Residual CNN with Focal Loss

Building on Model 1, we introduce: (i) residual (skip) connections in Blocks 2–4 using 1×1 convolutions for dimension matching, and (ii) Focal Loss as the training objective. L2 regularization ($\lambda = 10^{-4}$) is applied to all convolutional layers. Total parameters: ≈1.3M.

Residual connections address vanishing gradients:

$$\mathbf{a}_i = f(\mathbf{a}_{i-1}) + \mathbf{a}_{i-1} \tag{2}$$

ensuring gradients flow directly through shortcut branches.

Focal Loss is defined as:

$$\mathcal{L}_{\text{FL}}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \tag{3}$$

where p_t is the model's estimated probability for the ground-truth class, $\gamma = 2.0$ is the focusing parameter, and $\alpha = 0.25$ is the class-balancing weight. Training uses 2-epoch linear LR warmup, ReduceLROnPlateau (factor 0.5, patience 3), and EarlyStopping on `val_pr_auc` (patience 7).

4.3 Model 3: CNN + GLCM Texture Fusion (Tandem Baseline)

Model 3 introduces Gray-Level Co-occurrence Matrix (GLCM) texture features fused with a CNN backbone. GLCM captures second-order spatial intensity statistics; we extract contrast, correlation, energy, homogeneity, and entropy. These encode periodic regularity and unnatural smoothness characteristic of synthesis artifacts but absent in natural images.

The fusion strategy concatenates CNN-extracted deep features with the normalized GLCM feature vector before the classification head. Implemented in PyTorch and trained with cross-entropy loss.

4.4 Model 4: CNN + GLCM + Patch Loss

Model 4 extends Model 3 with a patch-level auxiliary loss. The image is divided into non-overlapping patches, each independently scored. The patch loss is:

$$\mathcal{L}_{\text{patch}} = \frac{1}{P} \sum_{p=1}^{P} \mathcal{L}_{\text{CE}}(f_p, y) \tag{4}$$

where P is the number of patches, f_p is the patch-level classifier output, and y is the image-level label propagated to all patches. The total loss combines image-level and patch-level cross-entropy terms.

4.5 Model 5: CNN + GLCM + Contrastive Loss

Model 5 replaces the patch auxiliary loss with a margin-based contrastive loss applied to patch embeddings:

$$\mathcal{L}_{\text{cont}} = Y \cdot d^2 + (1 - Y) \cdot \max(\alpha - d, 0)^2 \tag{5}$$

where d is Euclidean distance between patch embedding pairs, $Y \in \{0, 1\}$ indicates whether two patches share the same label (same class pulled together; different classes pushed apart with margin α). The combined loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_c \mathcal{L}_{\text{cont}} \tag{6}$$

5 Experiments and Results

5.1 Evaluation Metrics

We report: Accuracy, Balanced Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, MCC, Specificity (TNR), Sensitivity (TPR), FPR, FNR, NPV, and Brier Score. Threshold optimization is performed post-hoc by sweeping $\tau \in [0.10, 0.95]$ (step 0.01) and selecting the F1-maximizing threshold on the validation set.

5.2 TensorFlow Models: Full Test Split Results

Table 2: Test results for Models 1 and 2 (n=18,048).

Metric	M1: Plain CNN	M2: Res+FL
Opt. Threshold	0.500	0.410
Accuracy	0.7168	0.7196
Precision	0.6080	0.6044
Recall	0.8149	**0.8589**
F1 Score	0.6964	**0.7095**
Balanced Acc	0.7333	**0.7431**
ROC-AUC	0.7978	**0.8075**
PR-AUC	0.6690	**0.6786**
MCC	0.4580	**0.4803**
Specificity	0.6517	0.6272
Sensitivity	0.8149	0.8589
FPR	0.3483	0.3728
FNR	0.1851	**0.1411**

Model 2 (Residual+FL) outperforms the plain baseline on all minority-class-sensitive metrics. Recall improves by +4.4 points, ROC-AUC by +0.97 points, Balanced Accuracy by +0.98 points, and MCC by +0.022. The reduction in FNR from 0.1851 to 0.1411 is particularly notable for authenticity-verification applications.

5.3 PyTorch Tandem Models: Sampled Evaluation Results

The GLCM Baseline (Model 3) achieves the best F1 (0.762) and ROC-AUC (0.791) among tandem variants. Models 4 and 5 show competitive recall but reduced precision due to undertraining; their MCC values (0.341, 0.331) vs. Model 3's 0.451 reflect incomplete convergence. Fully trained versions are expected to substantially close this gap.

5.4 Loss Function Comparison

To illustrate why focal loss outperforms BCE under imbalance, consider a batch with 100 easy fake images ($p_t = 0.99$) and 1 hard real image ($p_t = 0.30$):

BCE: $\mathcal{L}_{\text{fakes}} \approx 460$, $\mathcal{L}_{\text{real}} \approx 1.2$ (383:1 ratio)

Focal Loss ($\gamma = 2$): $\mathcal{L}_{\text{fakes}} \approx 4.6$, $\mathcal{L}_{\text{real}} \approx 0.588$ (8:1 ratio)

The $(1 - p_t)^2$ modulating factor reduces the overwhelming dominance of easy majority-class samples by a factor of $\sim$48$\times$, enabling the minority class to provide meaningful gradient signal.

5.5 Threshold Tuning

A universal finding across all five models is that the F1-optimal threshold lies below the conventional 0.5 cutoff. Model 2 (Focal Loss) requires the smallest adjustment (0.41 vs. 0.50), indicating better-calibrated output probabilities compared to plain BCE. Model 3's near-0.5 optimal threshold (0.502) further suggests that richer feature representations naturally produce better-calibrated predictions.

6 Grad-CAM Interpretability

We apply Grad-CAM to the final convolutional layer of Models 1 and 2, analyzing all 8 convolutional layers. Spatial activation patterns show a consistent progression:

- **Early layers (Blocks 1–2):** Broad activations on edges and low-level texture patterns; not class-discriminative.
- **Middle layers (Block 3):** Differential activation begins; real images show stronger activation in specific semantic regions.
- **Deep layers (Block 4):** Highly selective; focus on generator-specific artifacts in AI images.

Model 2 (Focal Loss) produces notably sparser, more semantically focused heatmaps — a direct consequence of focal loss forcing discriminative rather than redundant feature learning.

7 Discussion

7.1 Positioning Against State of the Art

Table 4: Qualitative comparison with related work.

Method	Backbone	Acc	AUROC
CNNDetect (2020)	ResNet-50	$\sim$85%	–
Ensemble TL (2024)	DenseNet	94%	–
Texture Fusion (2025)	Custom 3-br.	–	>0.85
PPL (2026)	ViT-based	–	>0.90
Ours (M2)	Custom CNN	72%	**0.808**
Ours (M3)	CNN+GLCM	71%	0.791

Our architectures achieve competitive ROC-AUC with $\sim$20$\times$ fewer parameters than ResNet-50 baselines. The GLCM fusion approach independently validates the concurrent multi-modal texture fusion work [8], and our contrastive approach aligns with PPL [9], supporting both lines of research from a lightweight CNN perspective.

Table 3: Test results for tandem PyTorch models (n=250 sampled). Models 4 and 5 are undertrained (5 and 3 epochs respectively).

Model	Thr.	Acc	Prec	Rec	F1	ROC-AUC	PR-AUC	MCC	Brier
M3: GLCM Baseline	0.502	0.712	0.661	0.898	**0.762**	0.791	0.768	0.451	0.199
M4: +Patch Loss†	0.519	0.664	0.631	0.828	0.716	0.738	0.713	0.341	0.219
M5: +Contrastive†	0.446	0.652	0.612	0.875	0.720	0.709	0.705	0.331	0.220

† Undertrained: M4 at 5 epochs, M5 at 3 epochs.

7.2 Limitations and Future Work

1. **Protocol mismatch:** TF and PyTorch models were evaluated on different split sizes; unified evaluation is needed.
2. **Undertraining:** Models 4–5 need complete training runs before fair assessment.
3. **Generator coverage:** Cross-generator generalization to unseen diffusion models (DALL-E 3, Flux) should be evaluated explicitly.
4. **Calibration:** All models would benefit from temperature scaling before production deployment.

Future work will: (i) complete training of Models 4–5 under unified protocol, (ii) extend GLCM fusion with LBP and Fourier spectral features, (iii) evaluate cross-generator generalization, and (iv) ensemble all five models.

8 Conclusion

We presented a systematic empirical study of five progressively complex CNN architectures for AI-generated image detection. Each architectural innovation — residual connections, focal loss, GLCM texture fusion, patch-level supervision, and contrastive loss — contributes distinct advantages. Our best fully-trained model achieves ROC-AUC 0.8075 and F1 0.7095 with only 1.3M parameters. The GLCM-fused tandem model achieves F1 0.762 in its evaluation setting, and patch-level and contrastive objectives show strong early promise pending complete training. We release both our dataset and evaluation code publicly, providing an extensible and reproducible benchmark for the AI image detection community.

References

[1] R. Rombach et al. High-resolution image synthesis with latent diffusion models. *CVPR*, 2022.

[2] ACM CACM. As good as a coin toss: Human detection of AI-generated content. *CACM*, 2025.

[3] J. Frank et al. Leveraging frequency analysis for deep fake image recognition. *ICML*, 2020.

[4] S.-Y. Wang et al. CNN-generated images are surprisingly easy to spot... for now. *CVPR*, 2020.

[5] R. Durall et al. Watch your up-convolution. *CVPR*, 2020.

[6] D. Gragnaniello et al. Are GAN generated images easy to detect? *ICME*, 2021.

[7] R. Corvi et al. On the detection of synthetic images generated by diffusion models. *ICASSP*, 2023.

[8] Anonymous. Multi-modal texture fusion network for detecting AI-generated images. *Sci. Reports*, 2025.

[9] Anonymous. Enhance AI-generated image detection via panoptic patch learning. *arXiv:2504.01396*, 2026.

[10] F. F. Niloy et al. CFL-Net: Image forgery localization using contrastive learning. *WACV*, 2023.

[11] T.-Y. Lin et al. Focal loss for dense object detection. *ICCV*, 2017.

[12] K. He et al. Deep residual learning for image recognition. *CVPR*, 2016.

[13] Anonymous. AI-generated image detection using deep learning techniques. *IJARSCT*, 2024.

[14] S. Dhoot. AI_vs_Real dataset. Hugging Face Hub, 2026. `https://huggingface.co/datasets/ShreyashDhoot/AI_vs_Real`